



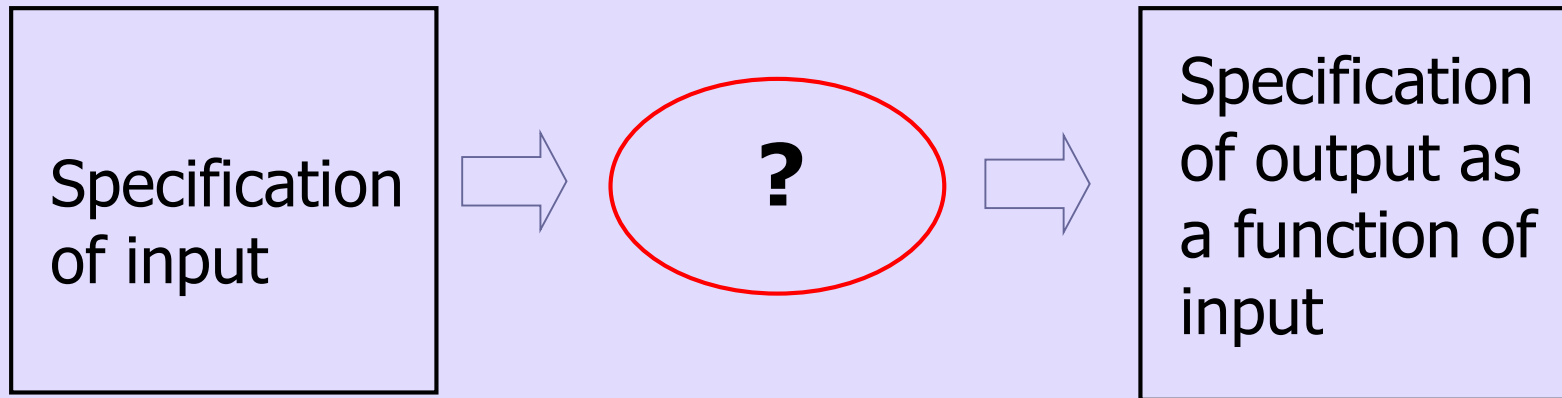
Asymptotic Analysis



Data Structures and Algorithms

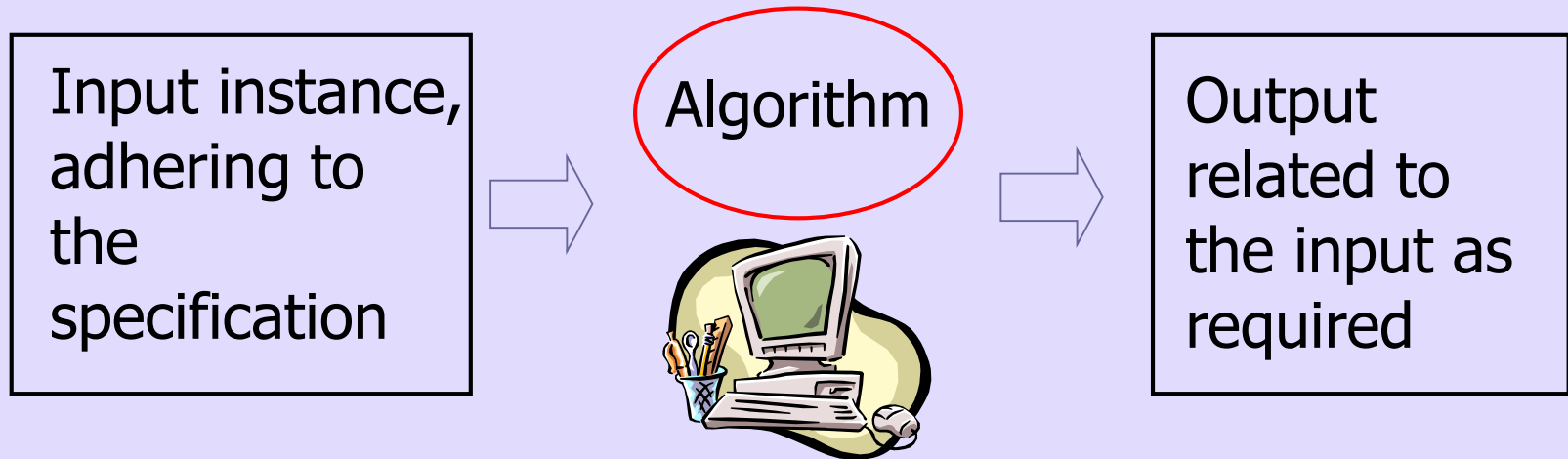
- **Algorithm:** Outline, the essence of a computational procedure, step-by-step instructions
- **Program:** an implementation of an algorithm in some programming language
- **Data structure:** Organization of data needed to solve the problem

Algorithmic problem



- Infinite number of input *instances* satisfying the specification.
 - For eg: A sorted, non-decreasing sequence of natural numbers of non-zero, finite length:
 - 1, 20, 908, 909, 100000, 10000000000.
 - 3.

Algorithmic Solution



- Algorithm describes actions on the input instance
- many correct algorithms for the same algorithmic problem

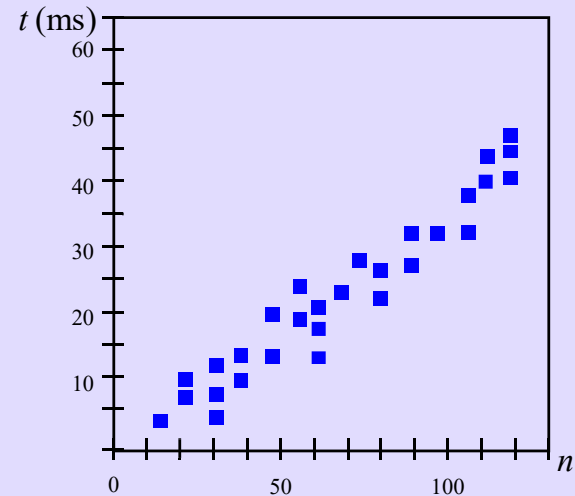


What is a Good Algorithm?

- Efficient:
 - Running time
 - Space used
- Efficiency as a function of input size:
 - The number of bits in an input number
 - Number of data elements (numbers, points)

Measuring the Running Time

How should we measure the running time of an **algorithm**?



Experimental Study

- ☐ Write a **program** that implements the algorithm
- ☐ Run the program with data sets of varying size and composition.
- ☐ Use a system call to get an accurate measure of the actual running time.



Limitations of Experimental Studies

- It is necessary to **implement** and test the algorithm in order to determine its running time.
- Experiments done only on a **limited set of inputs**,
 - may not be indicative of the running time on other inputs not included in the experiment.
- In order to compare two algorithms, the same **hardware and software environments** needed

Beyond Experimental Studies

We will develop a **general methodology** for analyzing running time of algorithms. This approach

- Uses a **high-level description** of the algorithm instead of testing one of its implementations.
- Takes into account **all possible inputs**.
- Allows one to evaluate the efficiency of any algorithm in a way that is **independent of the hardware and software environment**.

Example

□ **Algorithm** arrayMax(A, n):

Input: An array A storing n integers.

Output: The maximum element in A.

Pseudo-code (Functional / Recursive)

algorithm arrayMax(A[0..n-1])

{

A[0]

if n=1

max(arrayMax(A[0..n-2]), A[n-1])

o.w.

}

Pseudo-Code (imperative)

- A mixture of natural language and high-level programming concepts that describes the main ideas behind a generic implementation of a data structure or algorithm.

- Eg: **algorithm** arrayMax(A, n):

Input: An array A storing n integers.

Output: The maximum element in A.

currentMax $\leftarrow$ A[0]

for i $\leftarrow$ 1 **to** n-1 **do**

if currentMax < A[i] **then** currentMax $\leftarrow$ A[i]

return currentMax

Pseudo-Code

It is more structured than usual prose but less formal than a programming language

□ Expressions:

- use standard mathematical symbols to describe numeric and boolean expressions
- use $\leftarrow$ for assignment (“=” in Java)
- use = for equality relationship (“==” in Java)

□ Method Declarations:

- **algorithm** name(param1, param2)

Pseudo Code

- Programming Constructs:
 - decision structures: **if ... then ... [else ...]**
 - while-loops: **while ... do**
 - repeat-loops: **repeat ... until ...**
 - for-loop: **for ... do**
 - array indexing: **A[i], A[i,j]**
- Methods:
 - calls: object method(args)
 - returns: **return** value

Analysis of Algorithms

- **Primitive Operation:** Low-level operation independent of programming language. Can be identified in pseudo-code. For eg:
 - Data movement (assign)
 - Control (branch, subroutine call, return)
 - arithmetic and logical operations (e.g. addition, comparison)
- By inspecting the pseudo-code, we can count the number of primitive operations executed by an algorithm.

Example: Sorting

INPUT

sequence of numbers

$a_1, a_2, a_3, \dots, a_n$

2 5 4 10 7



OUTPUT

a permutation of the sequence of numbers

$b_1, b_2, b_3, \dots, b_n$

2 4 5 7 10

Correctness (requirements for the output)

For any given input the algorithm halts with the output:

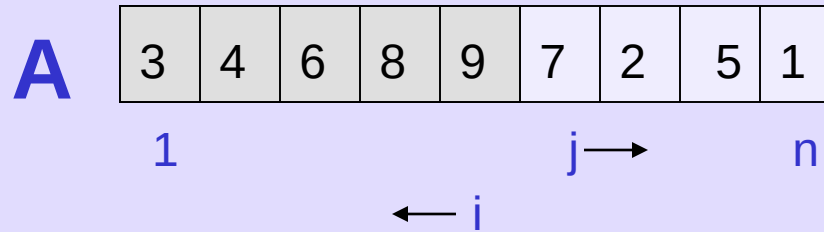
- $b_1 < b_2 < b_3 < \dots < b_n$
- $b_1, b_2, b_3, \dots, b_n$ is a permutation of $a_1, a_2, a_3, \dots, a_n$

Running time

Depends on

- number of elements (n)
- how (partially) sorted they are
- algorithm

Insertion Sort



Strategy

- Start “empty handed”
- Insert a card in the right position of the already sorted hand
- Continue until all cards are inserted/sorted

INPUT: $A[0..n-1]$ – an array of integers
OUTPUT: a permutation of A such that
 $A[0] \leq A[1] \leq \dots \leq A[n-1]$

Pseudo-code (Functional / Recursive)

algorithm insertionSort($A[0..n-1]$)

 $\{$

A[0] if n=1

insert(insertionSort(A[0..n-2]), A[n-1]) o.w.

}

algorithm insert($A[0..n-1]$, key)

{

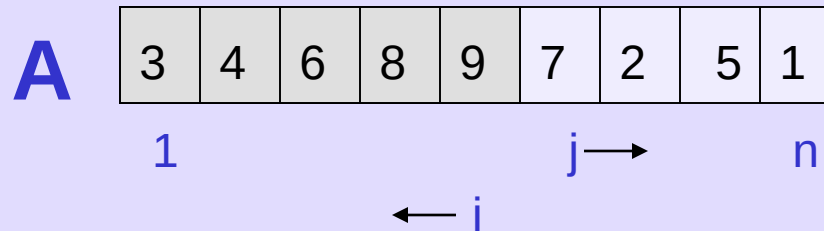
```
append(A[0..n-1], key)    if key ≥ A[n-1]
```

```
append(newarray(key), A[0])    if n=1&key<A[0]
```

```
append(insert(A[0..n-2],key), A[n-1])  o.w.
```

}

Insertion Sort



Strategy

- Start “empty handed”
- Insert a card in the right position of the already sorted hand
- Continue until all cards are inserted/sorted

INPUT: $A[0..n-1]$ – an array of integers
OUTPUT: a permutation of A such that $A[0] \leq A[1] \leq \dots \leq A[n-1]$

```
for j ← 1 to n-1 do
    key ← A[j]
    //insert A[j] into the sorted sequence
    A[0..j-1]
    i ← j-1
    while i ≥ 0 and A[i] > key
        do A[i+1] ← A[i]
           i--
    A[i+1] ← key
```

Analysis of Insertion Sort

	cost	Times
for $j \leftarrow 1$ to $n-1$ do	C_1	n
$\text{key} \leftarrow A[j]$	C_2	$n-1$
//insert $A[j]$ into the sorted sequence $A[0..j-1]$	0	$n-1$
$i \leftarrow j-1$	C_3	$n-1$
while $i \geq 0$ and $A[i] > \text{key}$	C_4	$\sum_{j=1}^{n-1} t_j$
do $A[i+1] \leftarrow A[i]$	C_5	$\sum_{j=1}^{n-1} (t_j - 1)$
$i--$	C_6	$\sum_{j=1}^{n-1} (t_j - 1)$
$A[i+1] \leftarrow \text{key}$	C_7	$n-1$

$$\begin{aligned} \text{Total time} = & n(C_1 + C_2 + C_3 + C_7) + \sum_{j=1}^{n-1} t_j (C_4 + C_5 + C_6) \\ & - (C_2 + C_3 + C_5 + C_6 + C_7) \end{aligned}$$

Best/Worst/Average Case

$$\text{Total time} = n(c_1 + c_2 + c_3 + c_7) + \sum_{j=1}^{n-1} t_j (c_4 + c_5 + c_6) - (c_2 + c_3 + c_5 + c_6 + c_7)$$

□ **Best case:**

- elements already sorted; $t_j=1$, running time = $f(n)$, i.e., *linear* time.

□ **Worst case:**

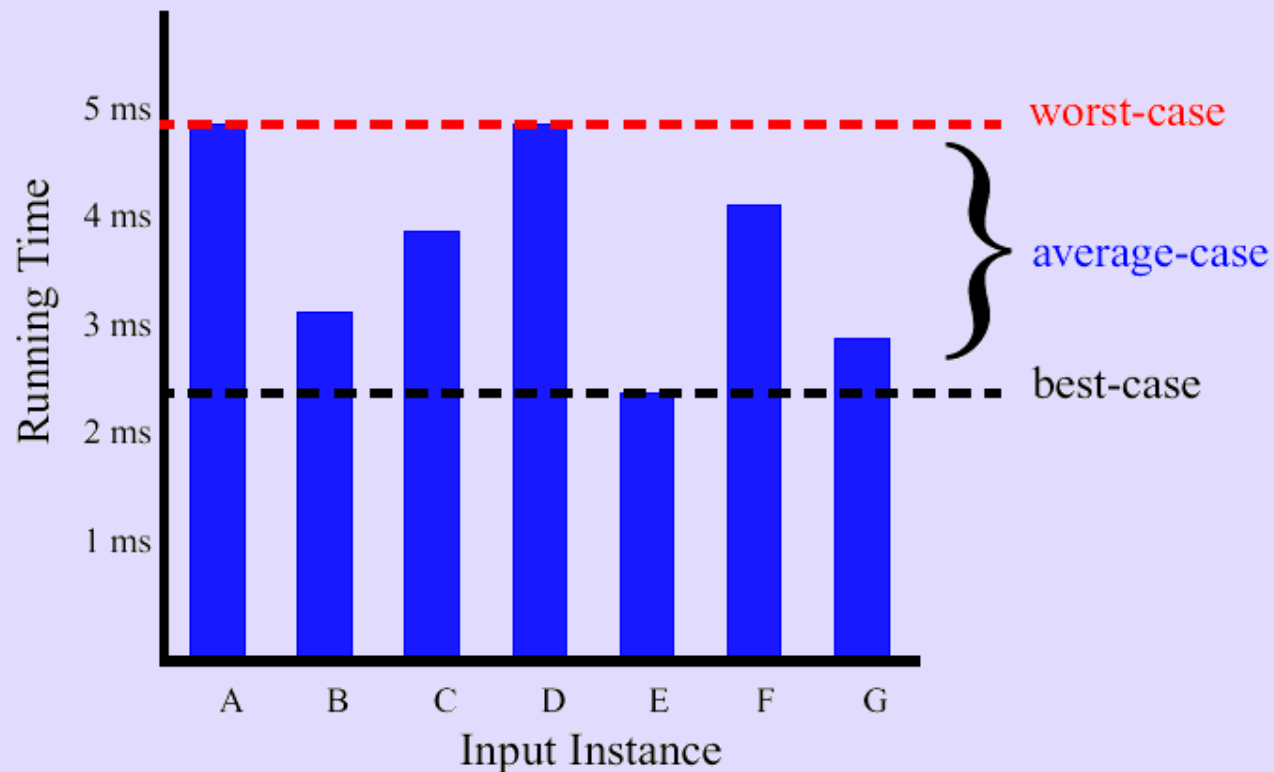
- elements are sorted in inverse order; $t_j=j+1$, running time = $f(n^2)$, i.e., *quadratic* time

□ **Average case:**

- $t_j=(j+1)/2$, running time = $f(n^2)$, i.e., *quadratic* time

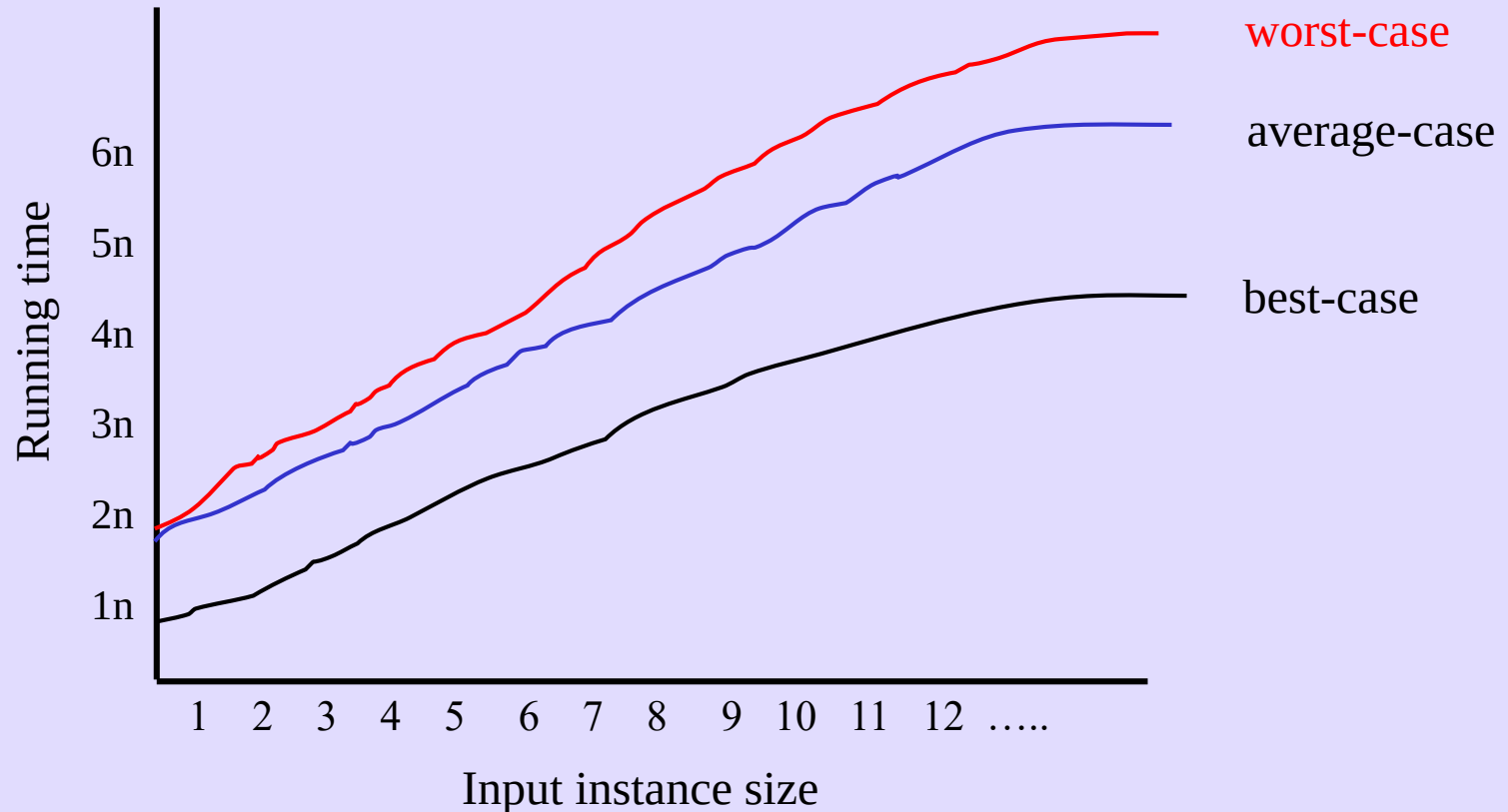
Best/Worst/Average Case (2)

- For a specific size of input n , investigate running times for different input instances:



Best/Worst/Average Case (3)

For inputs of all sizes:



Best/Worst/Average Case (4)

- **Worst case** is usually used: It is an upper-bound and in certain application domains (e.g., air traffic control, surgery) knowing the **worst-case** time complexity is of crucial importance
- For some algos **worst case** occurs fairly often
- **Average case** is often as bad as **worst case**
- Finding **average case** can be very difficult

Asymptotic Analysis

- Goal: to simplify analysis of running time by getting rid of "details", which may be affected by specific implementation and hardware
 - like "rounding": $1,000,001 \approx 1,000,000$
 - $3n^2 \approx n^2$
- Capturing the essence: how the running time of an algorithm increases with the size of the input *in the limit*.
 - Asymptotically more efficient algorithms are best for all but small inputs

Asymptotic Notation

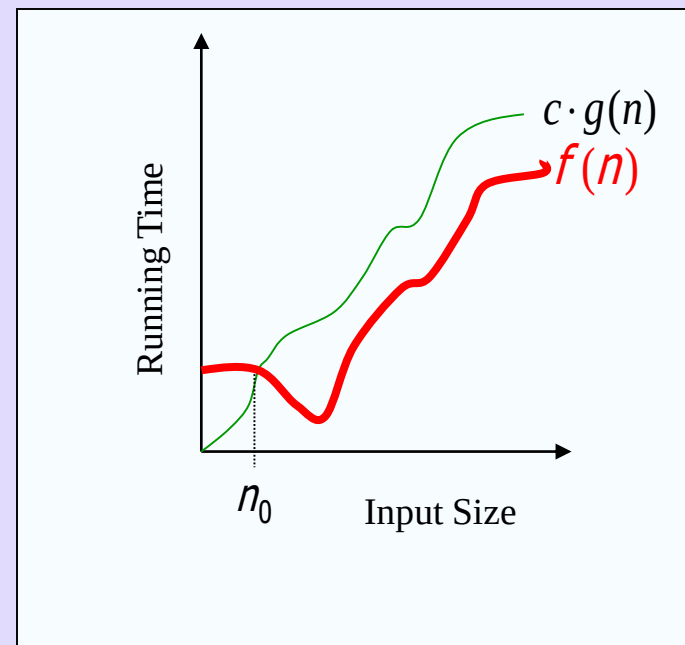
□ The “big-Oh” O-Notation

- asymptotic upper bound

- $f(n)$ is $O(g(n))$, if there exists constants c and n_0 , s.t. **$f(n) \leq c \cdot g(n)$** for all $n \geq n_0$

- $f(n)$ and $g(n)$ are functions over non-negative integers

- Used for *worst-case* analysis

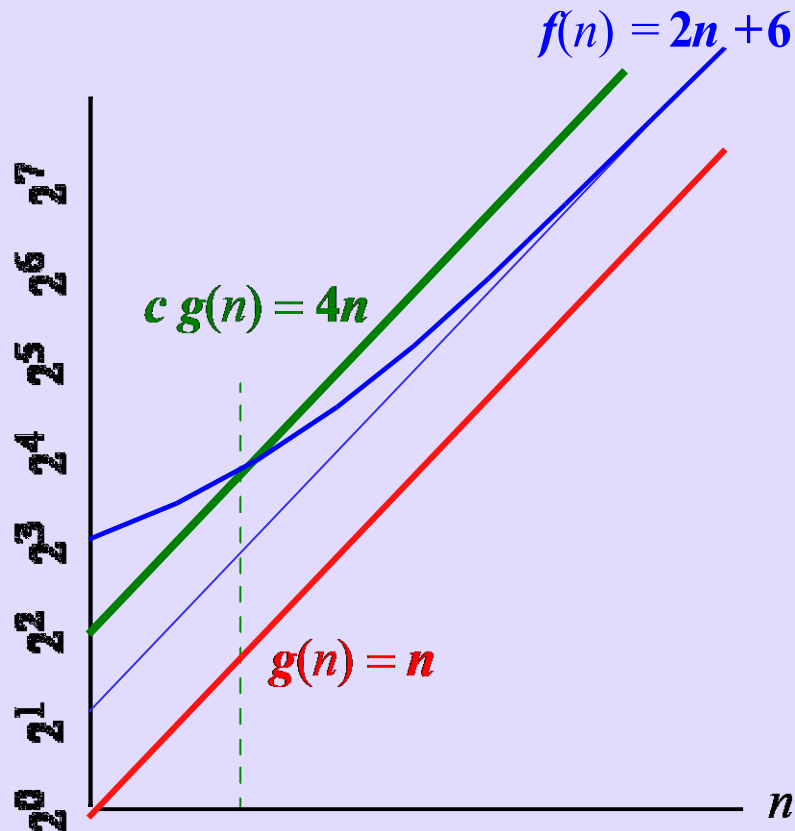


Example

For functions $f(n)$ and $g(n)$ there are positive constants c and n_0 such that: $f(n) \leq c g(n)$ for $n \geq n_0$

conclusion:

$2n+6$ is $O(n)$.



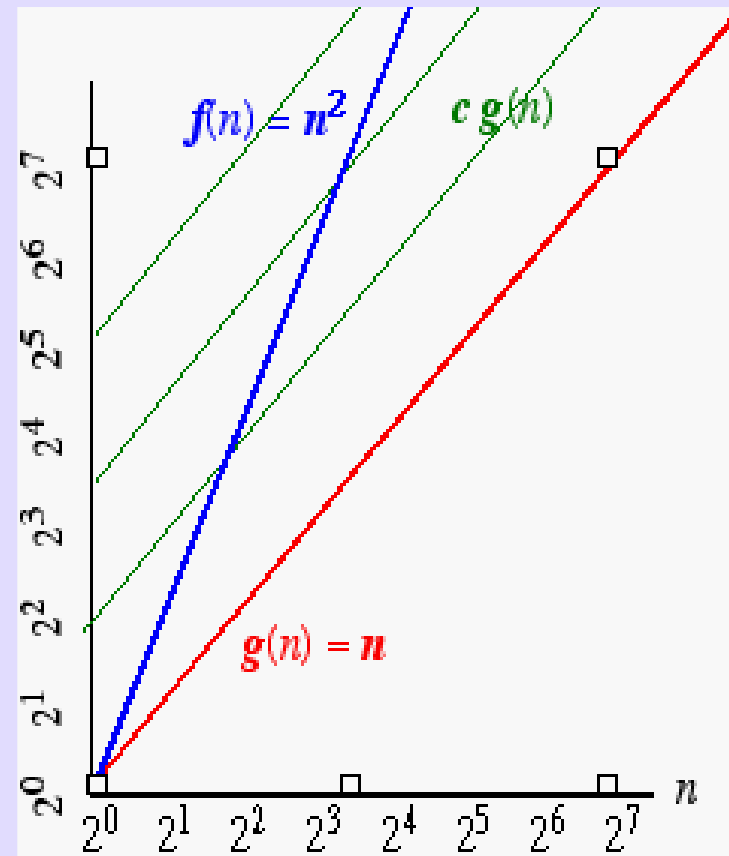
Another Example

On the other hand...

n^2 is not $O(n)$ because there is no c and n_0 such that:

$$n^2 \leq cn \text{ for } n \geq n_0$$

The graph to the right illustrates that no matter how large a c is chosen there is an n big enough that $n^2 > cn$).



Asymptotic Notation

- Simple Rule: Drop lower order terms and constant factors.
 - $50 n \log n$ is $O(n \log n)$
 - $7n - 3$ is $O(n)$
 - $8n^2 \log n + 5n^2 + n$ is $O(n^2 \log n)$
- Note: Even though $(50 n \log n)$ is $O(n^5)$, it is expected that such an approximation be of as small an order as possible

Asymptotic Analysis of Running Time

- Use O -notation to express number of primitive operations executed as function of input size.
- Comparing asymptotic running times
 - an algorithm that runs in $O(n)$ time is better than one that runs in $O(n^2)$ time
 - similarly, $O(\log n)$ is better than $O(n)$
 - hierarchy of functions: $\log n < n < n^2 < n^3 < 2^n$
- **Caution!** Beware of very large constant factors. An algorithm running in time $1,000,000 n$ is still $O(n)$ but might be less efficient than one running in time $2n^2$, which is $O(n^2)$

Example of Asymptotic Analysis

Algorithm prefixAverages1(X):

Input: An n -element array X of numbers.

Output: An n -element array A of numbers such that $A[i]$ is the average of elements $X[0], \dots, X[i]$.

for $i \leftarrow 0$ **to** $n-1$ **do**

$a \leftarrow 0$

for $j \leftarrow 0$ **to** i **do**

$a \leftarrow a + X[j]$ $\longleftarrow$ 1 step

$A[i] \leftarrow a/(i+1)$

return array A

Analysis: running time is $O(n^2)$

$\left. \begin{array}{l} \text{ } \end{array} \right\} n \text{ iterations}$
 $\left. \begin{array}{l} i \text{ iterations} \\ \text{with} \\ i=0,1,2,\dots,n-1 \end{array} \right\}$

A Better Algorithm

Algorithm prefixAverages2(X):

Input: An n -element array X of numbers.

Output: An n -element array A of numbers such that $A[i]$ is the average of elements $X[0], \dots, X[i]$.

$s \leftarrow 0$

for $i \leftarrow 0$ **to** n **do**

$s \leftarrow s + X[i]$

$A[i] \leftarrow s/(i+1)$

return array A

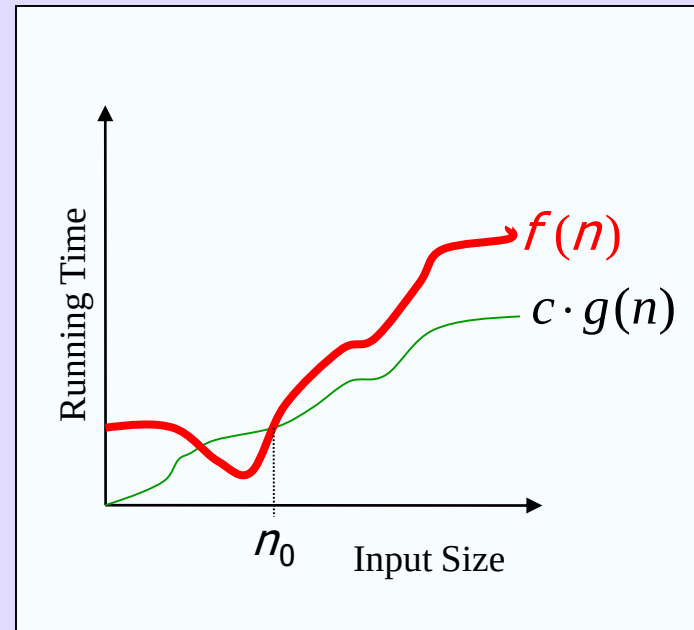
Analysis: Running time is $O(n)$

Asymptotic Notation (*terminology*)

- Special classes of algorithms:
 - **Logarithmic**: $O(\log n)$
 - **Linear**: $O(n)$
 - **Quadratic**: $O(n^2)$
 - **Polynomial**: $O(n^k)$, $k \geq 1$
 - **Exponential**: $O(a^n)$, $a > 1$
- “Relatives” of the Big-Oh
 - $\Omega(f(n))$: **Big Omega** -asymptotic *lower* bound
 - $\Theta(f(n))$: **Big Theta** -asymptotic *tight* bound

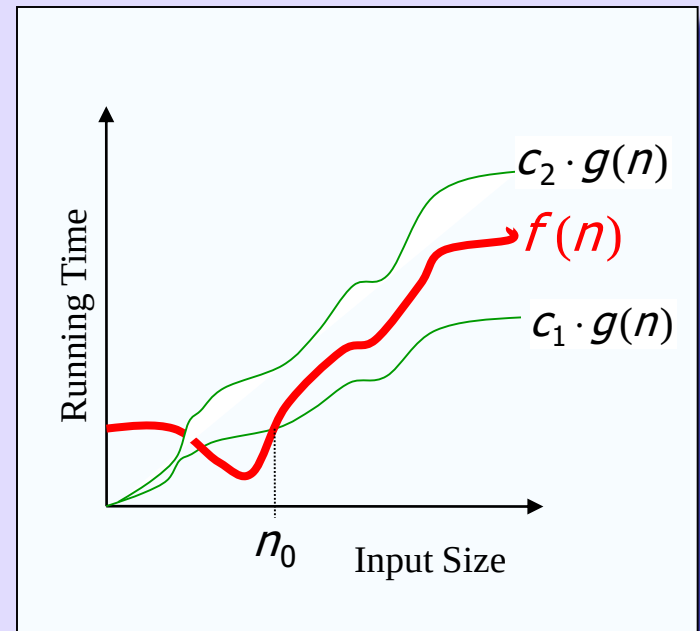
Asymptotic Notation

- The “big-Omega” Ω -Notation
 - asymptotic lower bound
 - $f(n)$ is $\Omega(g(n))$ if there exists constants c and n_0 , s.t.
 $c \cdot g(n) \leq f(n)$ for $n \geq n_0$
- Used to describe *best-case* running times or lower bounds for algorithmic problems
 - E.g., lower-bound for searching in an unsorted array is $\Omega(n)$.



Asymptotic Notation

- The “big-Theta” Θ –Notation
 - asymptotically tight bound
 - $f(n)$ is $\Theta(g(n))$ if there exists constants c_1 , c_2 , and n_0 , s.t. $\mathbf{c_1 \, g(n) \leq f(n) \leq c_2 \, g(n)}$ for $n \geq n_0$
- $f(n)$ is $\Theta(g(n))$ if and only if $f(n)$ is $O(g(n))$ and $f(n)$ is $\Omega(g(n))$
- $O(f(n))$ is often misused instead of $\Theta(f(n))$



Asymptotic Notation

Two more asymptotic notations

- "Little-Oh" notation $f(n)$ is $o(g(n))$
non-tight analogue of Big-Oh
 - For every $c > 0$, there should exist n_0 , s.t.
 $f(n) \leq c g(n)$ for $n \geq n_0$
 - Used for **comparisons** of running times.
If $f(n)$ is $o(g(n))$, it is said that $g(n)$
dominates $f(n)$.
- "Little-omega" notation $f(n)$ is $\omega(g(n))$
non-tight analogue of Big-Omega

Asymptotic Notation

□ Analogy with real numbers

$$\square f(n) \text{ is } O(g(n)) \quad \cong \quad f \leq g$$

$$\square f(n) \text{ is } \Omega(g(n)) \quad \cong \quad f \geq g$$

$$\square f(n) \text{ is } \Theta(g(n)) \quad \cong \quad f = g$$

$$\square f(n) \text{ is } o(g(n)) \quad \cong \quad f < g$$

$$\square f(n) \text{ is } \omega(g(n)) \quad \cong \quad f > g$$

□ Abuse of notation: $f(n) = O(g(n))$ actually means $f(n) \in O(g(n))$

Comparison of Running Times

Running Time	Maximum problem size (n)		
	1 second	1 minute	1 hour
$400n$	2500	150000	9000000
$20n \log n$	4096	166666	7826087
$2n^2$	707	5477	42426
n^4	31	88	244
2^n	19	25	31